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Final Business Report – Credit Card Default Prediction

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Introduction

Problem Statement:
The objective is to predict the probability of credit card default using customer account, transaction, and delinquency information. This is a classic supervised classification problem, critical for credit risk management.

Business Need:
Predicting default risk helps financial institutions:

- Reduce credit losses
- Optimize customer acquisition strategies
- Ensure regulatory compliance with capital adequacy

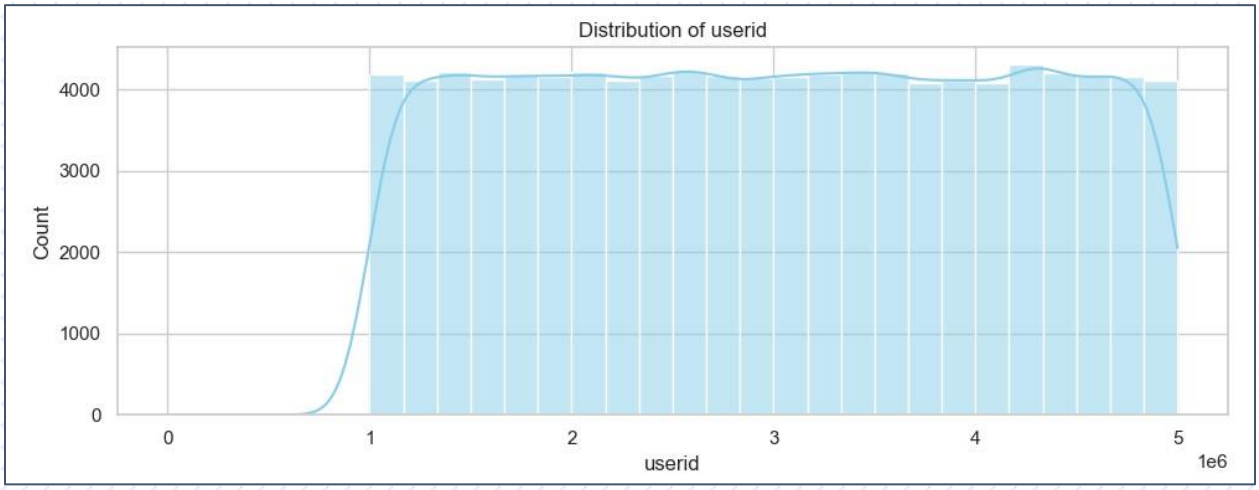
2. EDA and Business Implementation

Data Summary:

- Total Observations: ~100,000
- Features: 35+
- Target Variable: default (0 = No, 1 = Yes)

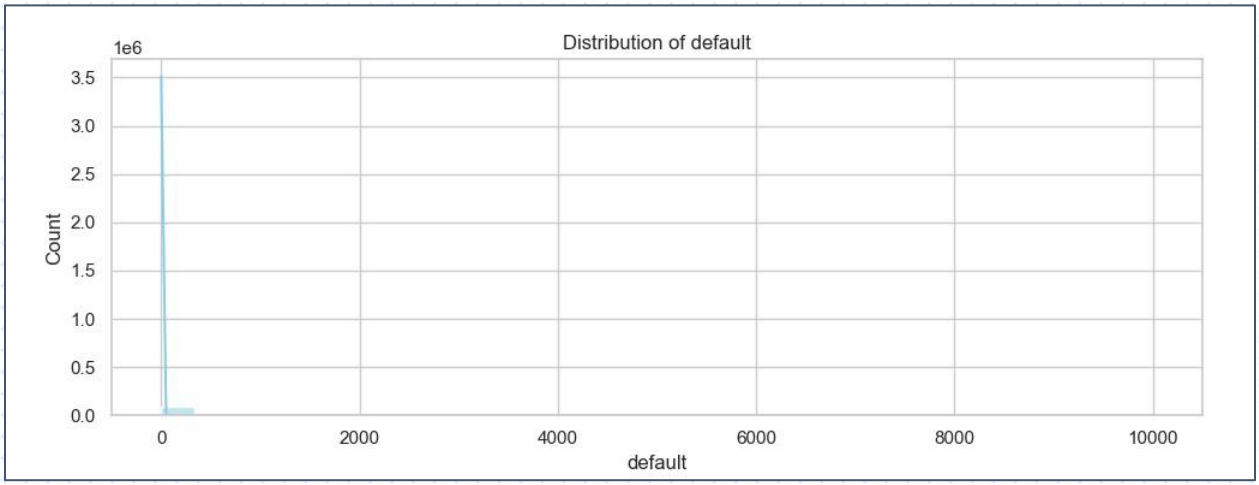
Univariate Analysis

Distribution of User IDs



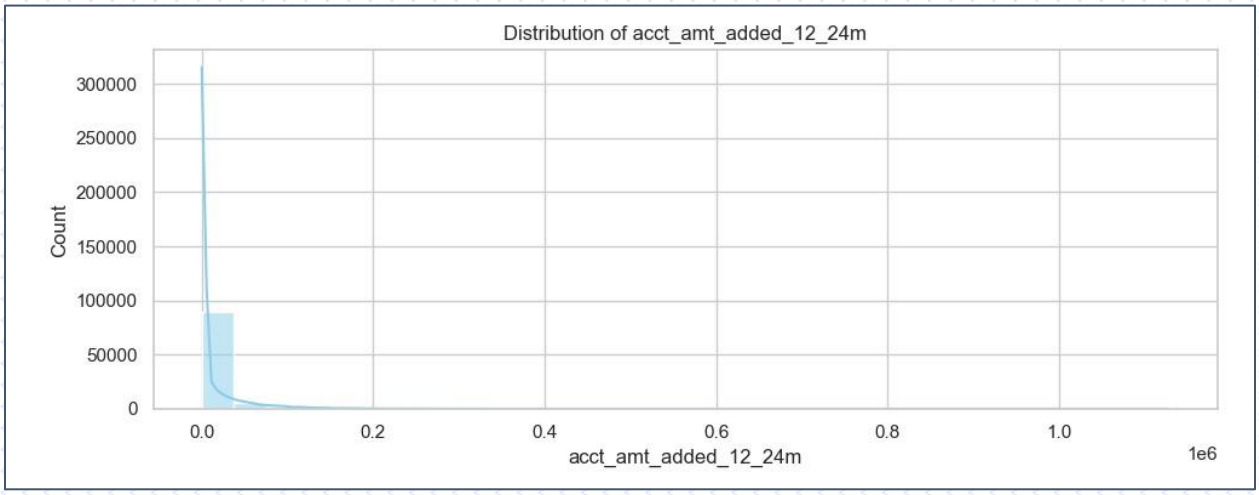
The histogram illustrates the distribution of user IDs, with the x-axis ranging from 0 to 5 million and the y-axis from 0 to 4000. The distribution is relatively uniform between 1 million and 5 million user IDs, with a sharp increase around 1 million, indicating a large number of users in that range. The histogram is smoothed with a curve, providing a visual representation of the data's probability density.

Distribution of Default



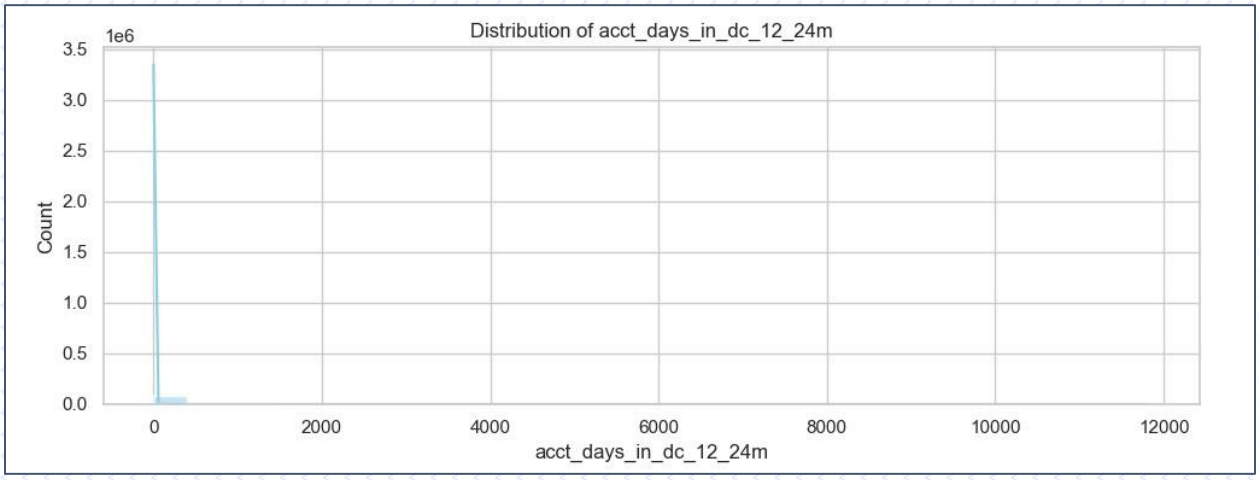
The histogram titled "Distribution of default" shows a high count for low default values, as indicated by a very tall bar at the beginning of the x-axis, which ranges from 0 to 10000. The y-axis ranges from 0 to 3.5 million, and the rest of the histogram is relatively flat, suggesting low counts for higher default values.

Distribution of acct_amt_added_12_24m



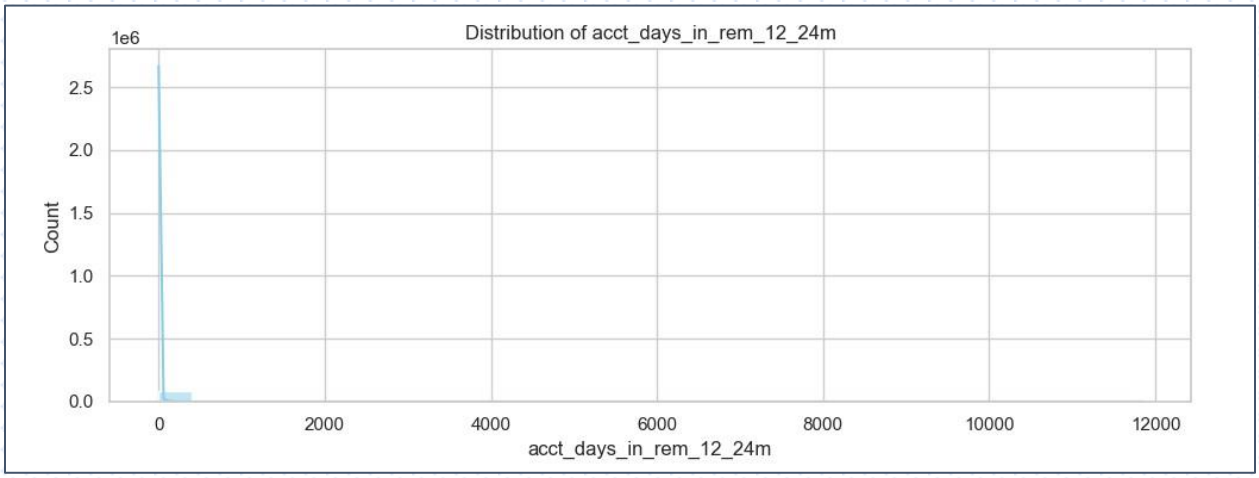
The histogram displays the distribution of "acct_amt_added_12_24m", with the x-axis ranging from 0.0 to 1 million and the y-axis from 0 to 300,000. There is a high concentration of values near 0.0, with a long tail extending to the right, indicating a large number of accounts with a small amount added. A curve is overlaid to show the estimated probability density function.

Distribution of acct_days_in_dc_12_24m



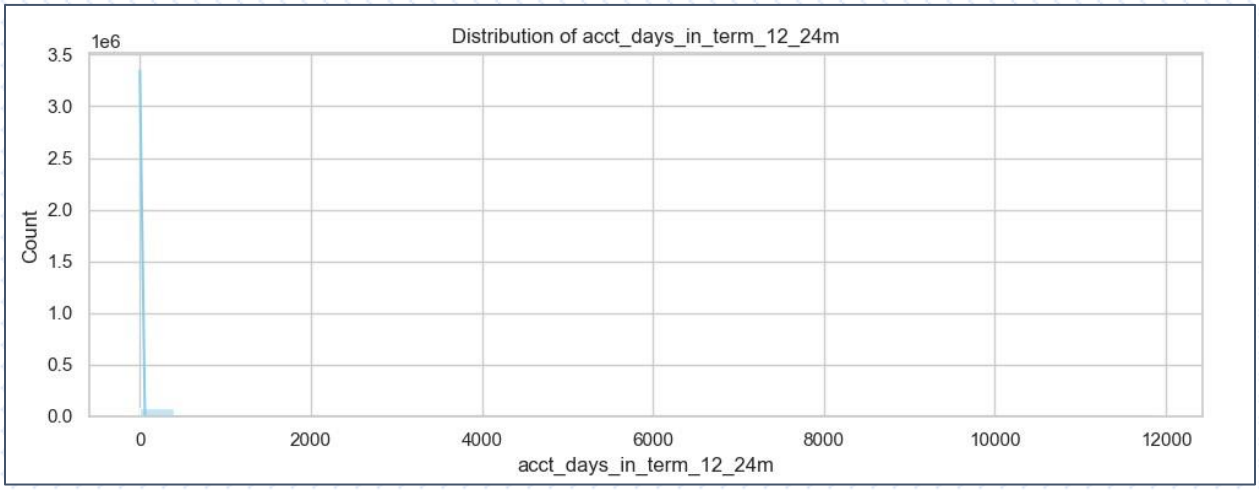
The histogram shows the distribution of "acct_days_in_dc_12_24m", with the x-axis ranging from 0 to 12000 and the y-axis from 0 to 3.5 million. The data is heavily skewed to the right, with a very high count near 0, indicating that most data points have a small value for "acct_days_in_dc_12_24m".

Distribution of acct_days_in_rem_12_24m



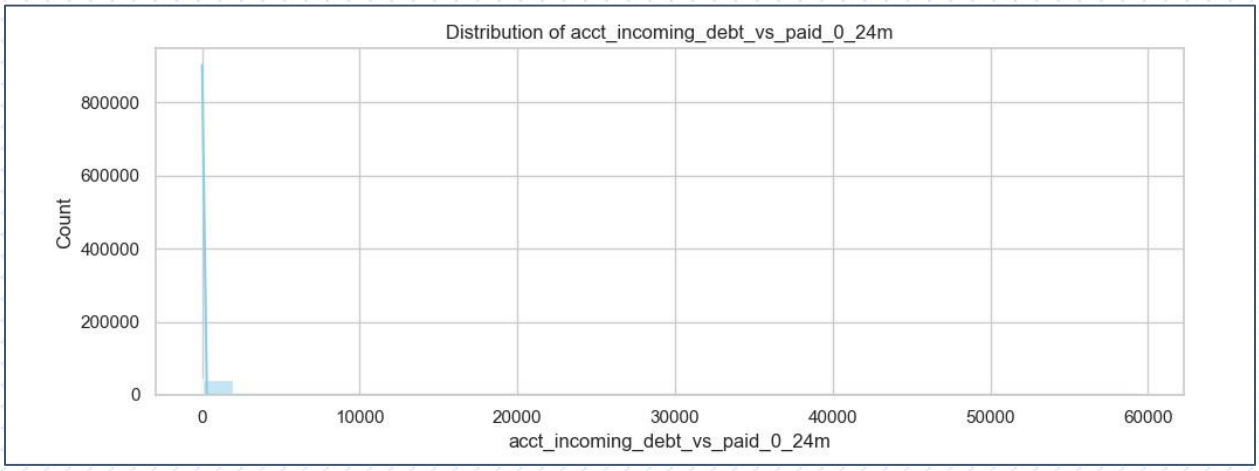
The histogram represents the distribution of "acct_days_in_rem_12_24m", with the x-axis ranging from 0 to 12000 and the y-axis from 0 to 2.5 million. It shows a highly skewed distribution, with a very large peak near 0 and a long tail extending to the right.

Distribution of acct_days_in_term_12_24m



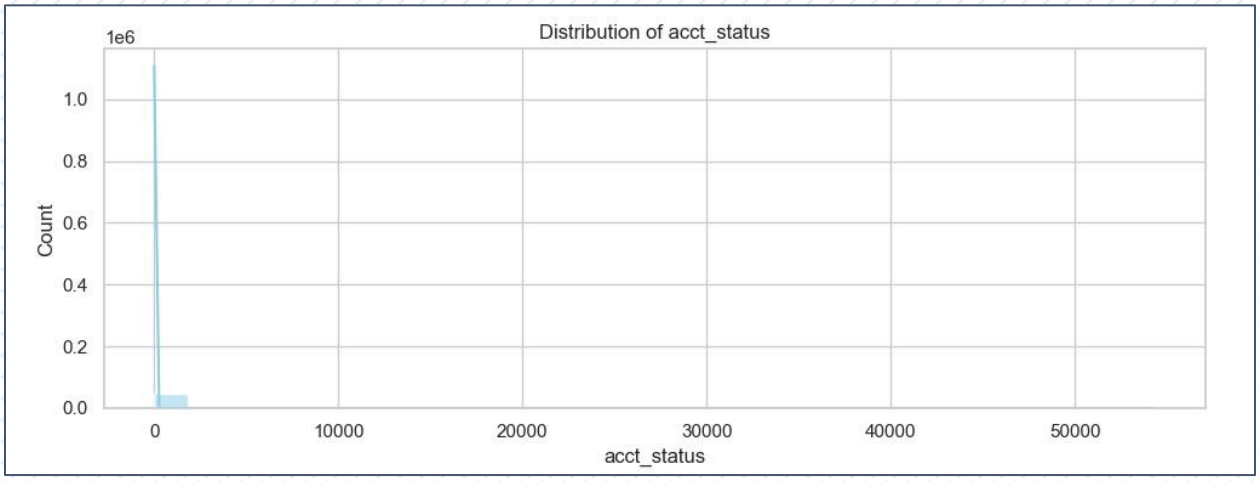
The histogram for "acct_days_in_term_12_24m" shows a very high count near 0, with the count rapidly decreasing as the value increases. The x-axis ranges from 0 to 12000, and the y-axis from 0 to 3.5 million.

Distribution of acct_incoming_debt_vs_paid_0_24m



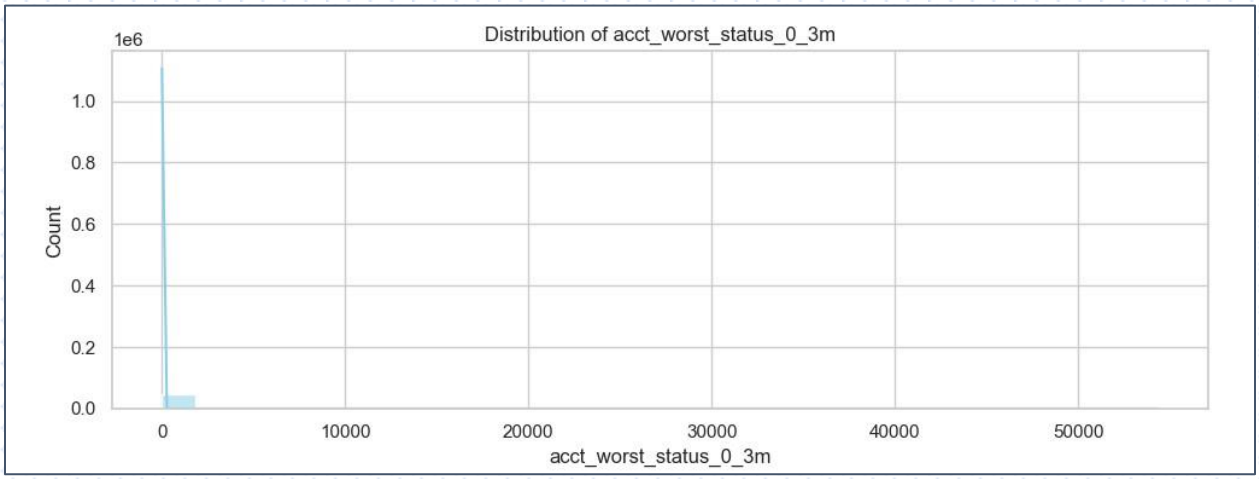
The histogram illustrates the distribution of "acct_incoming_debt_vs_paid_0_24m", with the x-axis ranging from 0 to 60000 and the y-axis from 0 to 800,000. There is a very high count near 0, with the count rapidly decreasing as the value increases.

Distribution of acct_status



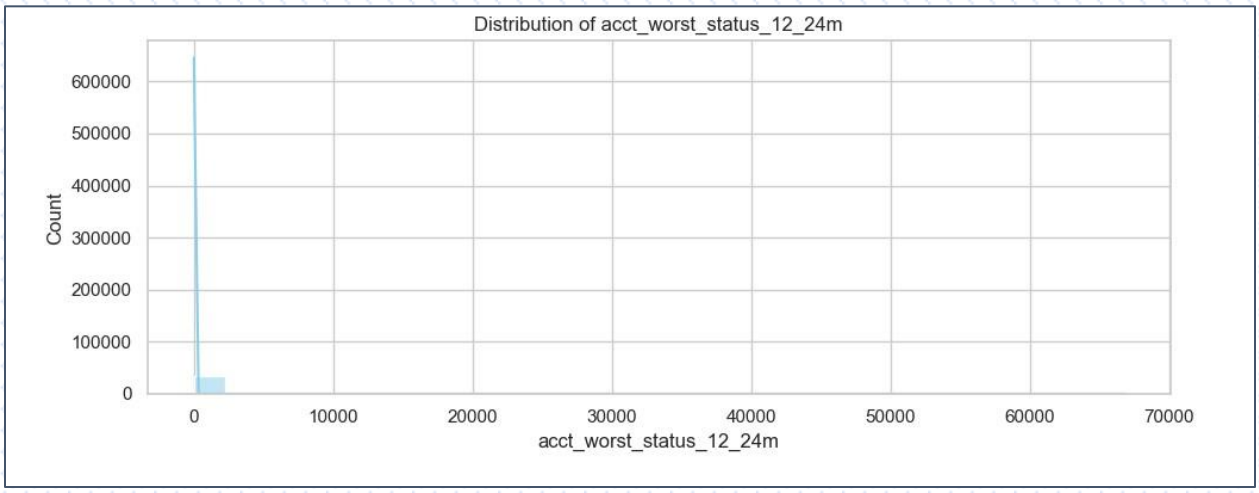
The histogram titled "Distribution of acct_status" shows a very tall bar near 0 on the x-axis, indicating a high count for that value. The x-axis ranges from 0 to 50000, and the y-axis from 0 to 1 million. The rest of the histogram is relatively empty, suggesting most data is concentrated at the lower end.

Distribution of acct_worst_status_0_3m



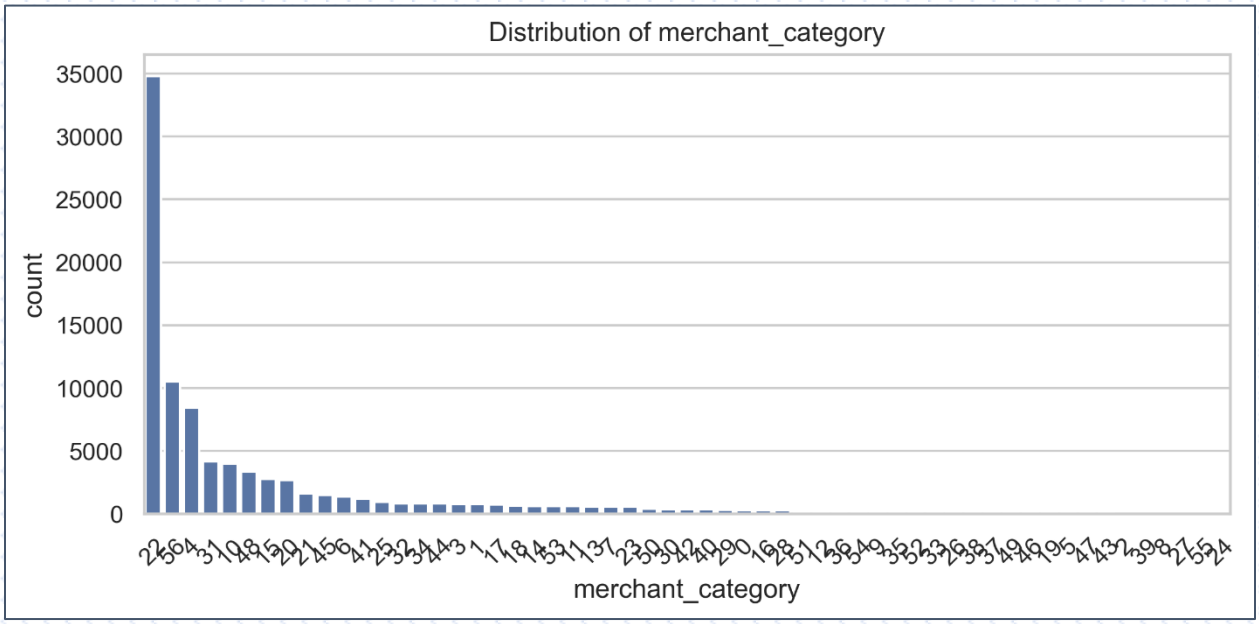
The histogram shows the distribution of "acct_worst_status_0_3m", with a very high peak at the beginning, close to 1 million, and then quickly decreasing. The x-axis ranges from 0 to 50000, and the y-axis from 0 to 1 million, indicating most data points have a low value.

Distribution of acct_worst_status_12_24m



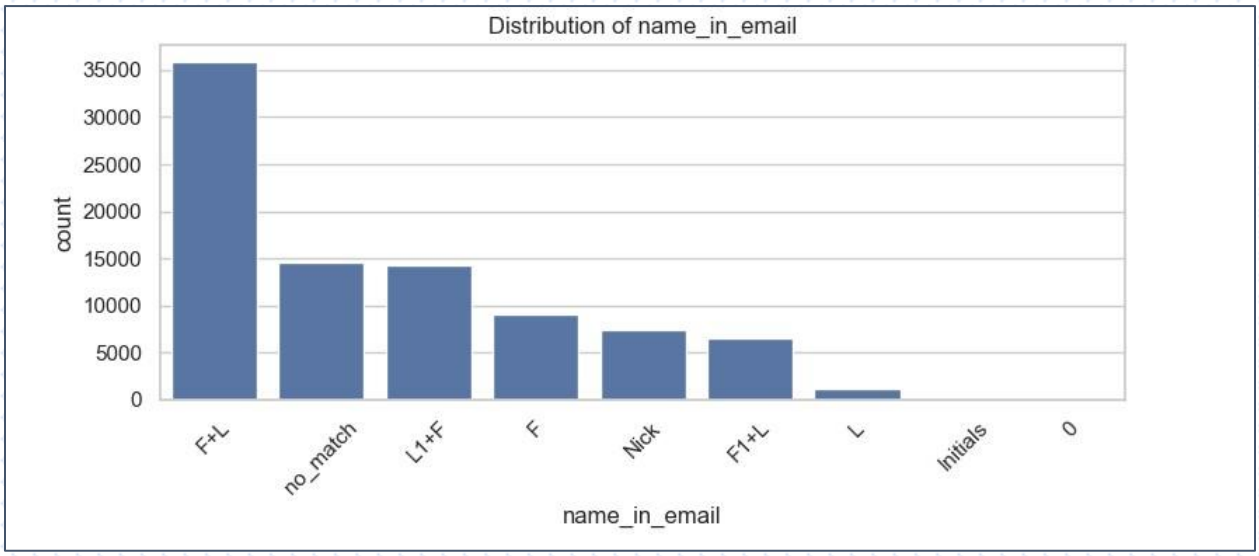
The histogram titled "Distribution of acct_worst_status_12_24m" shows a very high count near 0, with the count rapidly decreasing as the x-axis value increases. The x-axis ranges from 0 to 70000, and the y-axis from 0 to 600,000, with a tall bar at the beginning followed by a much smaller bar.

Distribution of Merchant Categories:



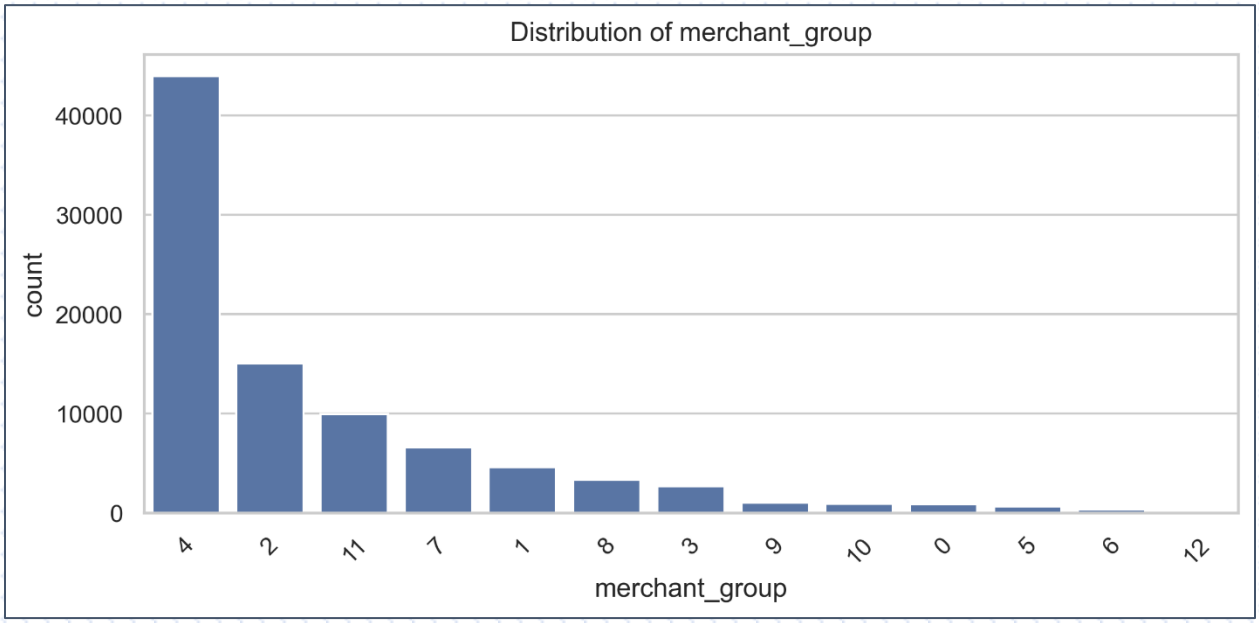
The bar chart shows that merchant category 2564 has the highest count, followed by 31. This suggests that these categories are the most frequently occurring among the merchant categories.

Distribution of Name in Email:



The bar chart shows the distribution of different categories of "name_in_email". The category F+L has the highest count, followed by no_match and L1+F. This indicates that these are the most common patterns in email names.

Distribution of Merchant Groups:



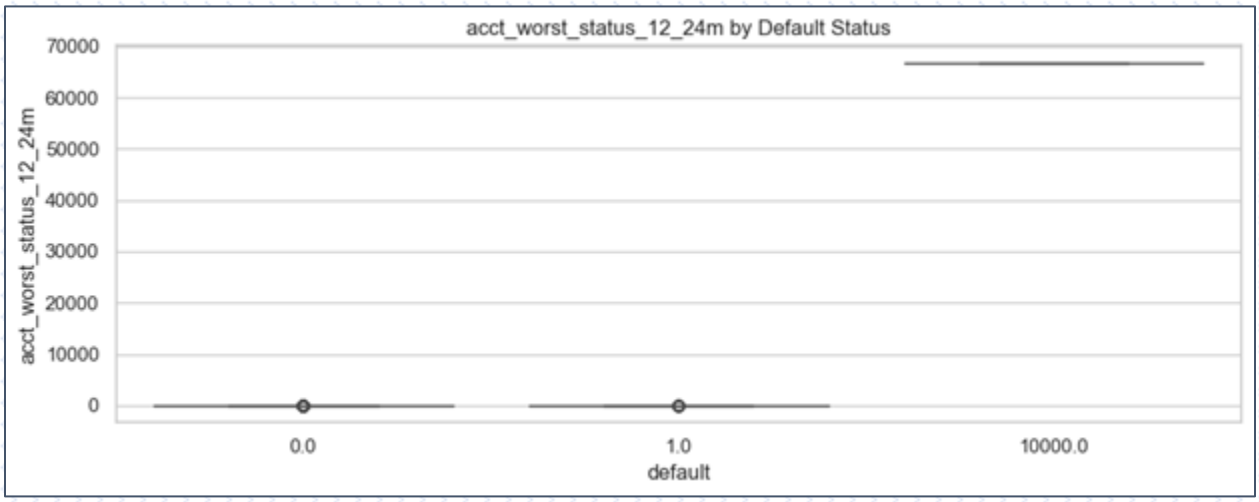
The bar chart indicates that merchant group '4' has the highest count, followed by '2' and '11'. This suggests that these groups are the most common among the merchant groups.

Key EDA Insights:

- Univariate: Features like age, recovery_debt, and worst_status showed high variability.

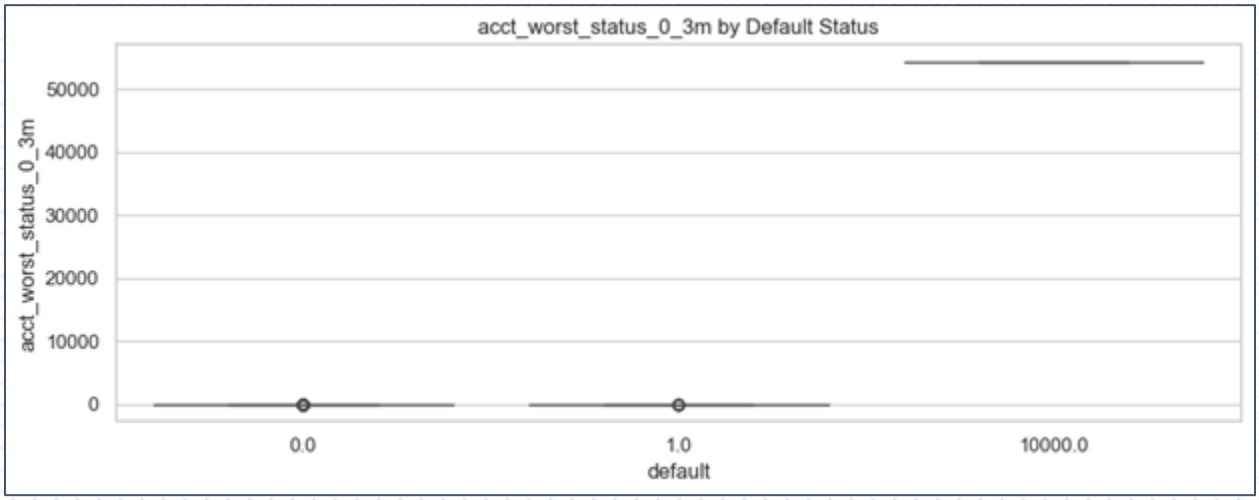
Bivariate Analysis:

Worst Account Status (12-24 months) by Default Status



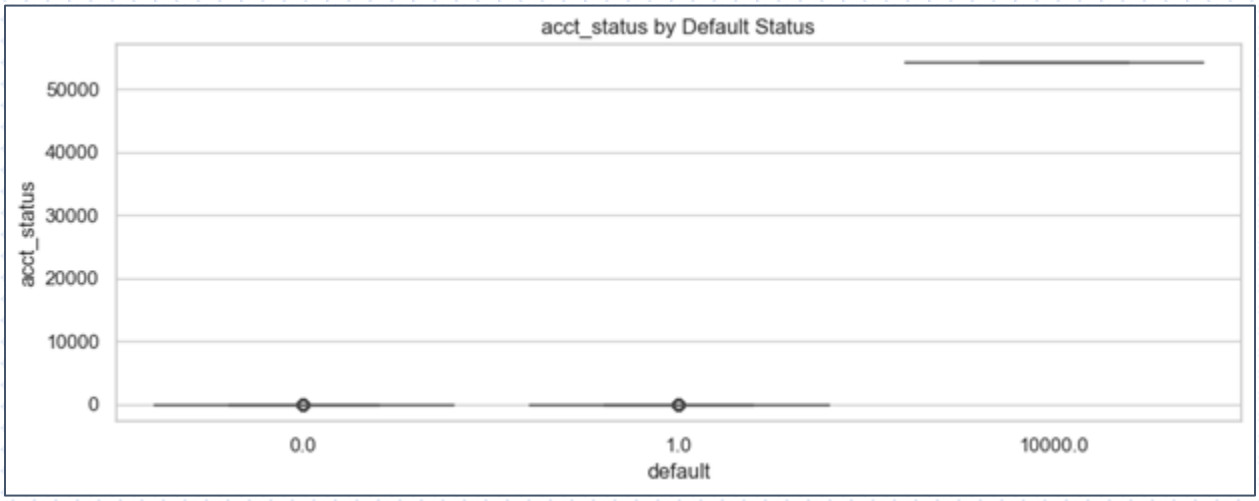
The boxplot that the categories 0.0 and 1.0 are very low, close to 0, while the category 10000.0 is much higher, around 66000. This suggests a significantly worse account status for the default status of 10000.0 over a longer period.

Worst Account Status (0-3 months) by Default Status



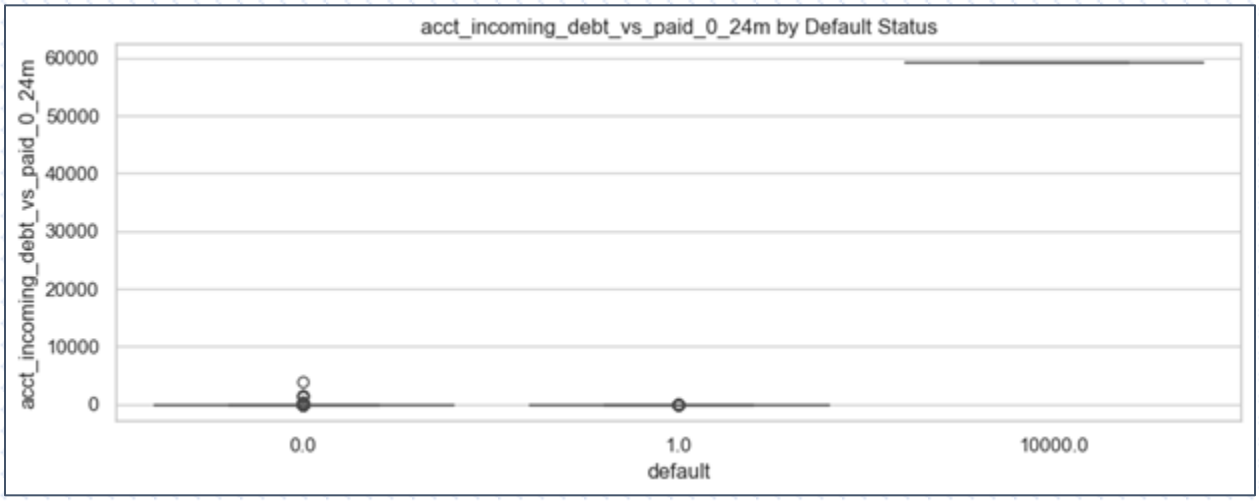
The boxplot on Page 19 shows that the categories 0.0 and 1.0 are near 0, while the category 10000.0 is around 50000. This indicates a much worse account status for the default status of 10000.0 over the short term.

Account Status by Default Status



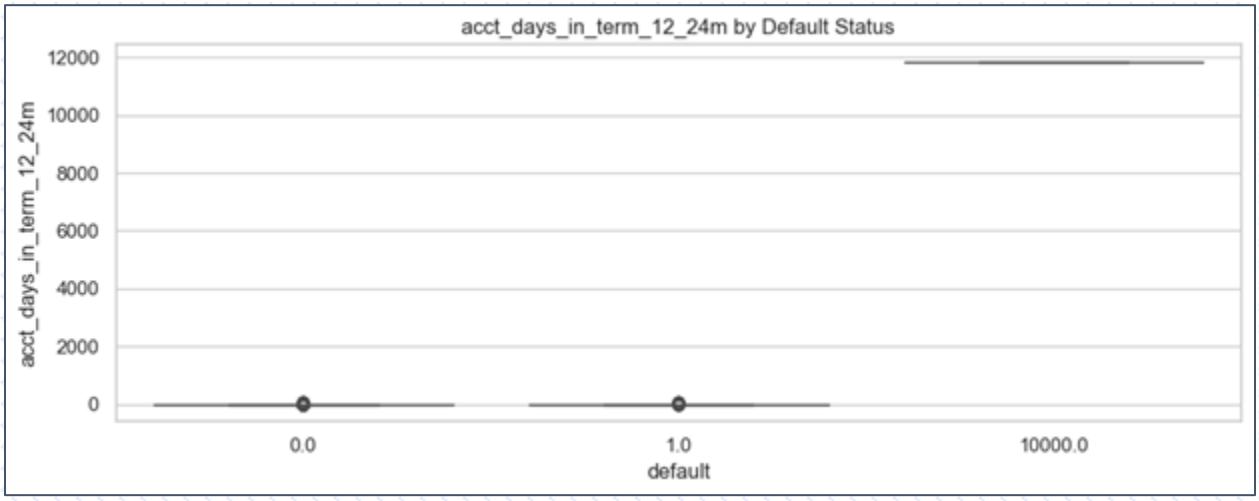
The boxplot indicates that the categories 0.0 and 1.0 are very close to 0, while the category 10000.0 is near the top of the y-axis. This suggests a significant difference in account status for the default status of 10000.0.

Incoming Debt vs. Paid (0-24 months) by Default Status



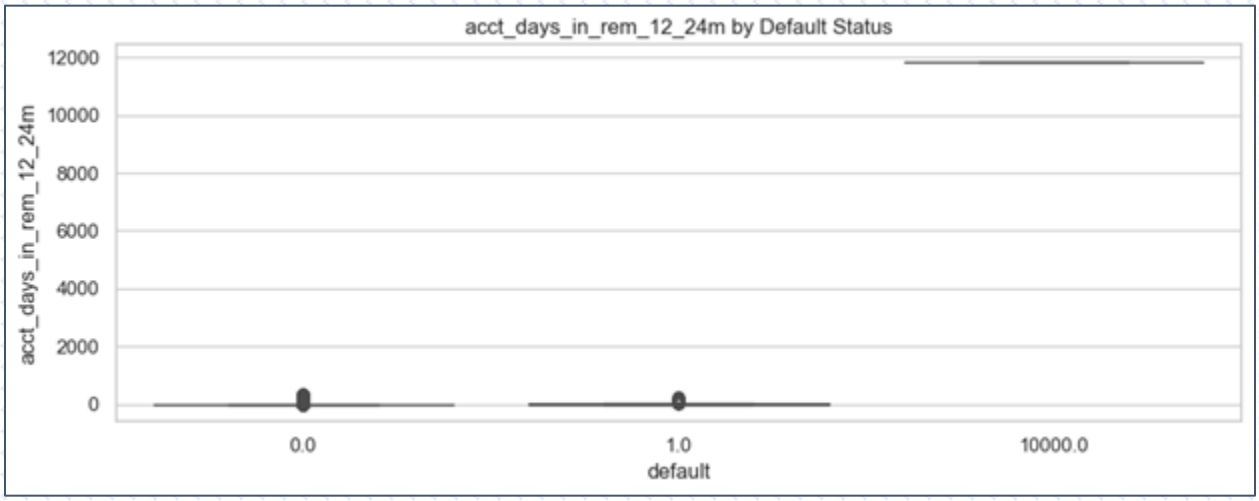
The boxplot shows that the categories 0.0 and 1.0 are concentrated near 0, with some outliers. The category 10000.0 is a single line at approximately 60000, indicating a high ratio of incoming debt to paid amounts for this default status.

Days in Term (12-24 months) by Default Status



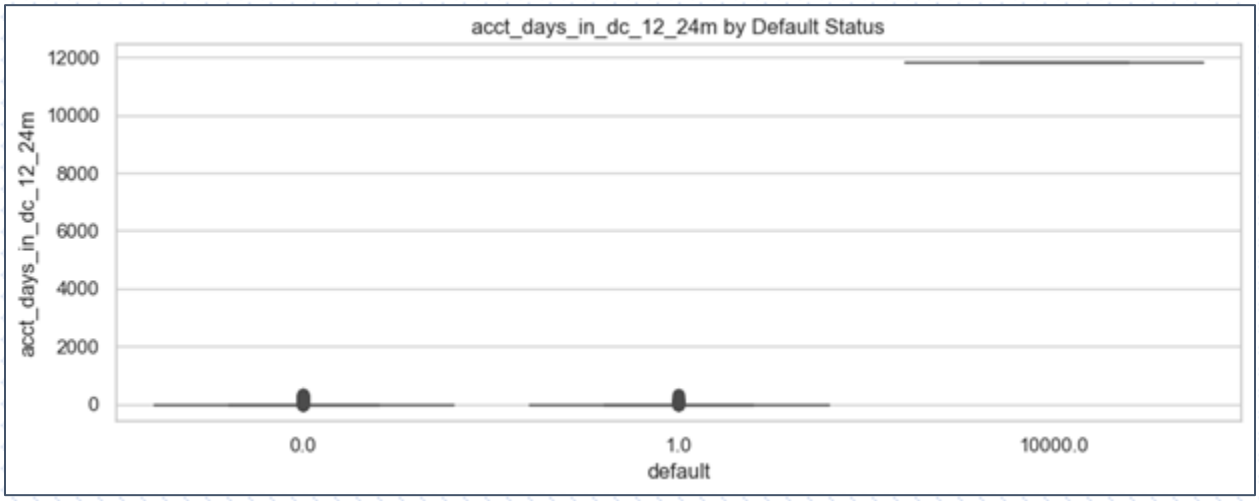
The boxplot reveals that the categories 0.0 and 1.0 are very close to 0, while the category 10000.0 is near 12000. This suggests that accounts with a default status of 10000.0 have longer terms.

Days in Remission (12-24 months) by Default Status



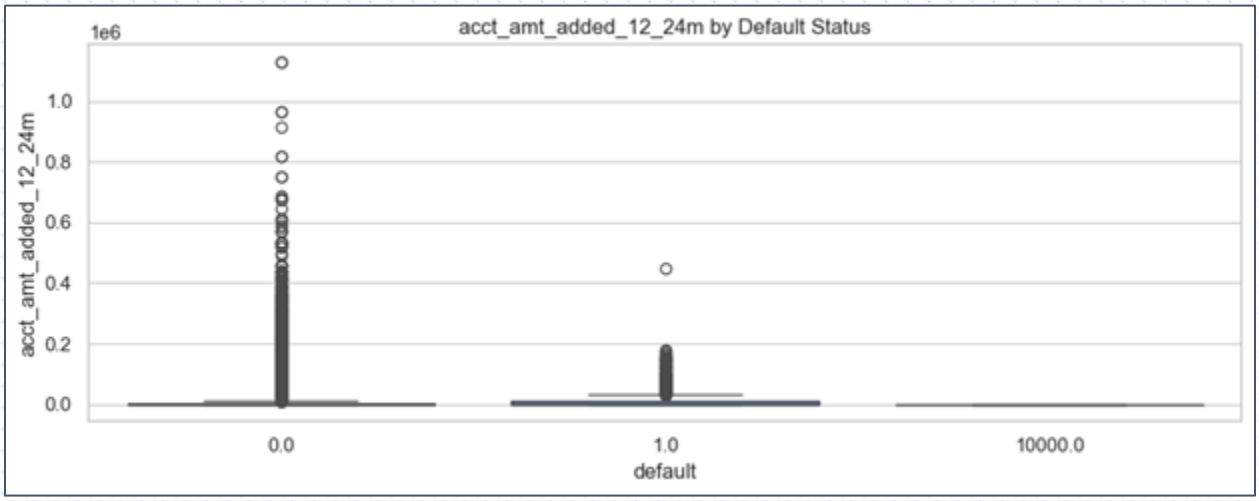
The boxplot shows that the categories 0.0 and 1.0 are near 0, while the category 10000.0 is near 12000. This indicates that accounts with a default status of 10000.0 also spend more days in remission.

Days in Debt Collection (12-24 months) by Default Status



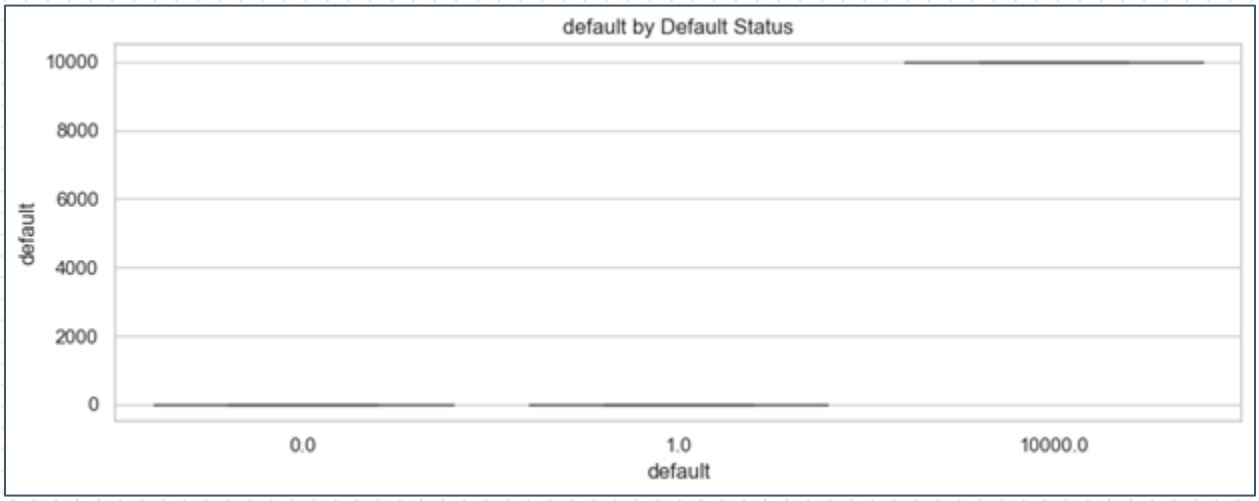
The boxplot indicates that the categories 0.0 and 1.0 have distributions near 0, while the category 10000.0 is near 12000. This suggests that accounts with a default status of 10000.0 spend significantly more days in debt collection.

Account Amount Added (12-24 months) by Default Status



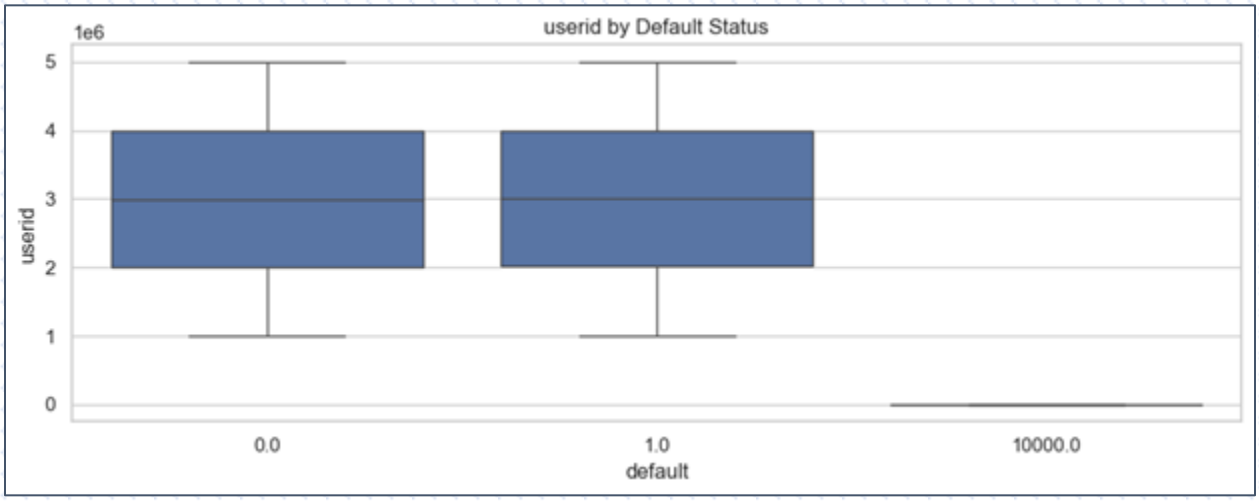
The boxplot shows the distribution of "acct_amt_added_12_24m" for each default status. The category 0.0 has many outliers above the box, while category 1.0 has one outlier. The category 10000.0 has no outliers, suggesting a more consistent distribution of account amounts added for this default status.

Default by Default Status



The bar graph illustrates the default values for each default status category. The bars for 0.0 and 1.0 are very short, close to 0, while the bar for 10000.0 reaches the top of the y-axis at 10000. This indicates that the default status of 10000.0 is associated with the highest default value.

User ID by Default Status

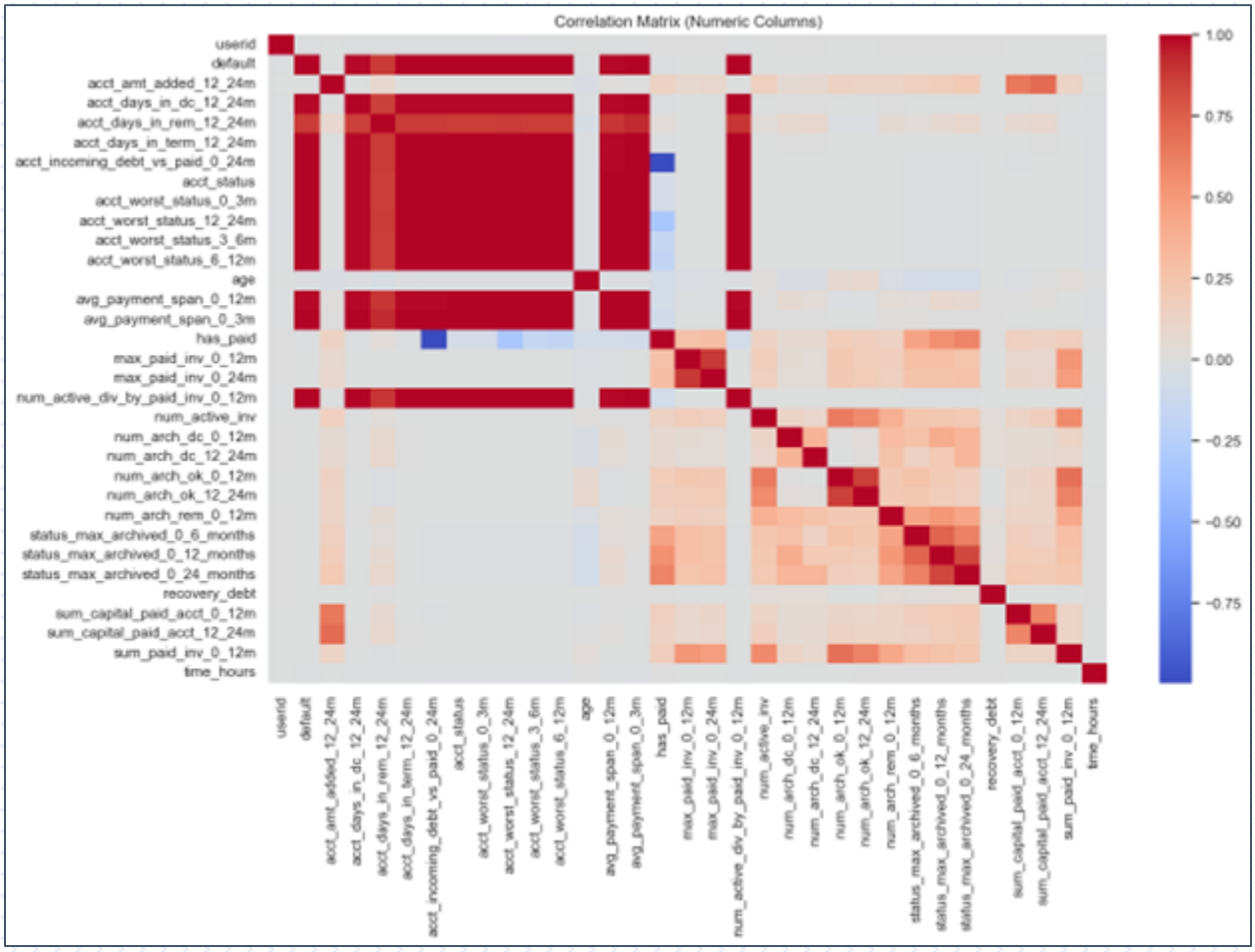


The boxplot shows the distribution of user IDs across different default statuses. The categories 0.0 and 1.0 have similar distributions, with medians around 3 million and ranges from approximately 1 million to 5 million. However, the category 10000.0 has a much lower distribution, with values close to 0, indicating a significant difference in user ID distribution for this default status.

Key EDA Insights:

- Bivariate: Strong relationship observed between worst_status_max, repayment_ratio, and default.

Multivariate Analysis:



Correlation Matrix Analysis

The correlation matrix visualizes the relationships between various numeric columns. Here's an analysis of the key insights:

1. **Strong Positive Correlations:**

- **`acct_days_in_dc_12_24m` and `acct_worst_status_12_24m`**: These variables show a strong positive correlation, suggesting that more days in debt collection are associated with worse account status.

- **`acct_days_in_rem_12_24m` and `acct_days_in_term_12_24m`**: These two are also highly correlated, indicating that days in remission often correspond with days in term.

2. **Strong Negative Correlations:**

- **`avg_payment_span_0_12m` and `avg_payment_span_0_3m`**: There is a noticeable negative correlation, suggesting inverse behavior in payment spans over different timeframes.

3. **Neutral to Weak Correlations:**

- Many variables show neutral (around zero) or weak correlations, indicating independent behavior or minimal linear relationship between those pairs.

4. **Clusters of Related Variables:**

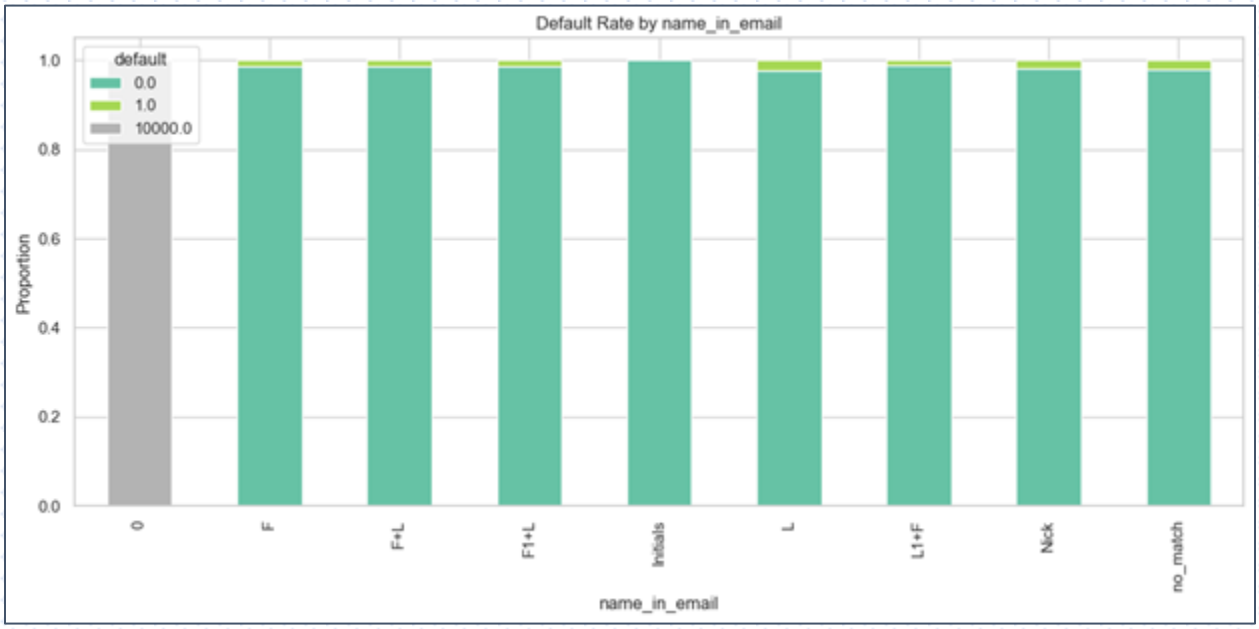
- Variables related to account status and days in various financial states (debt collection, remission, term) tend to cluster together, reflecting their interconnected nature in financial assessments.

5. **Interpretation for Decision-Making:**

- Understanding these correlations can help in risk assessment and targeting interventions. For example, monitoring days in debt collection could predict account status issues, allowing for preemptive action.

This matrix provides valuable insights into how different financial metrics interact, which can guide strategic financial decisions and risk management.

Default Rate by Name in Email Analysis



1. **Overview**:

- The bar chart shows the default rates across different patterns of names in email addresses, categorized by default statuses: 0.0, 1.0, and 10000.0.

2. **Default Status 0.0**:

- The majority of email name patterns have a high proportion of transactions with a default status of 0.0, indicating successful completion and low default rates.

3. **Default Status 1.0**:

- A small fraction of defaults falls into the 1.0 category across all email name patterns. This suggests occasional minor issues that are typically resolved.

4. **Default Status 10000.0**:

- This status is rare across all name patterns, indicating minimal severe defaults. The presence of 10000.0 is negligible, showing consistent transaction reliability regardless of email name pattern.

5. **Pattern-Specific Insights:**

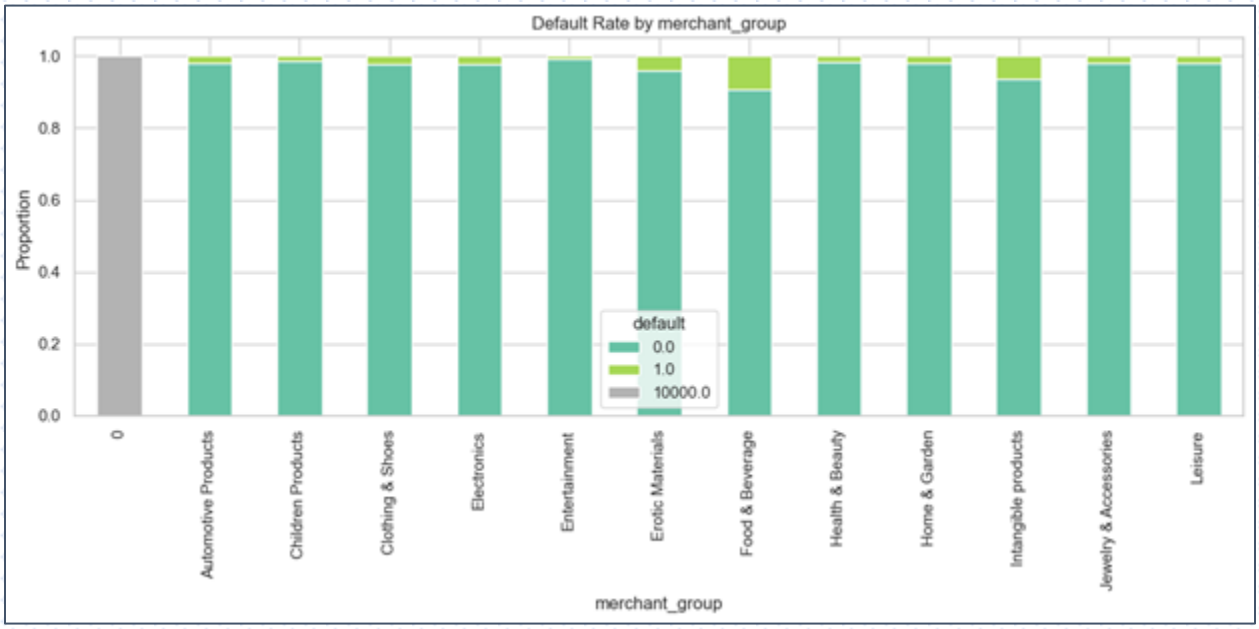
- There is no significant variation in default rates across different name patterns in emails, suggesting that the structure or pattern of an email name does not heavily influence default behavior.

6. **Strategic Implications:**

- Since there are no significant differences, email name patterns may not need focused risk management. Efforts can be directed towards other factors that might influence default rates.

This analysis indicates that email name patterns have minimal impact on default rates, maintaining consistent transaction success across patterns.

Default Rate by Merchant Group Analysis



1. **Overview**:

- The bar chart displays default rates across different merchant groups, categorized into default statuses: 0.0, 1.0, and 10000.0.

2. **Default Status 0.0**:

- A significant majority of transactions in all merchant groups have a default status of 0.0, indicating low default rates and successful transaction completions.

3. **Default Status 1.0**:

- There is a small proportion of transactions with a default status of 1.0 across most merchant groups. This suggests occasional, minor issues that are generally resolved.

4. **Default Status 10000.0**:

- This status is rare but visible in certain groups, indicating a potential for severe defaults. Groups with noticeable bars for this status might require special attention to manage risks.

5. **Group-Specific Insights:**

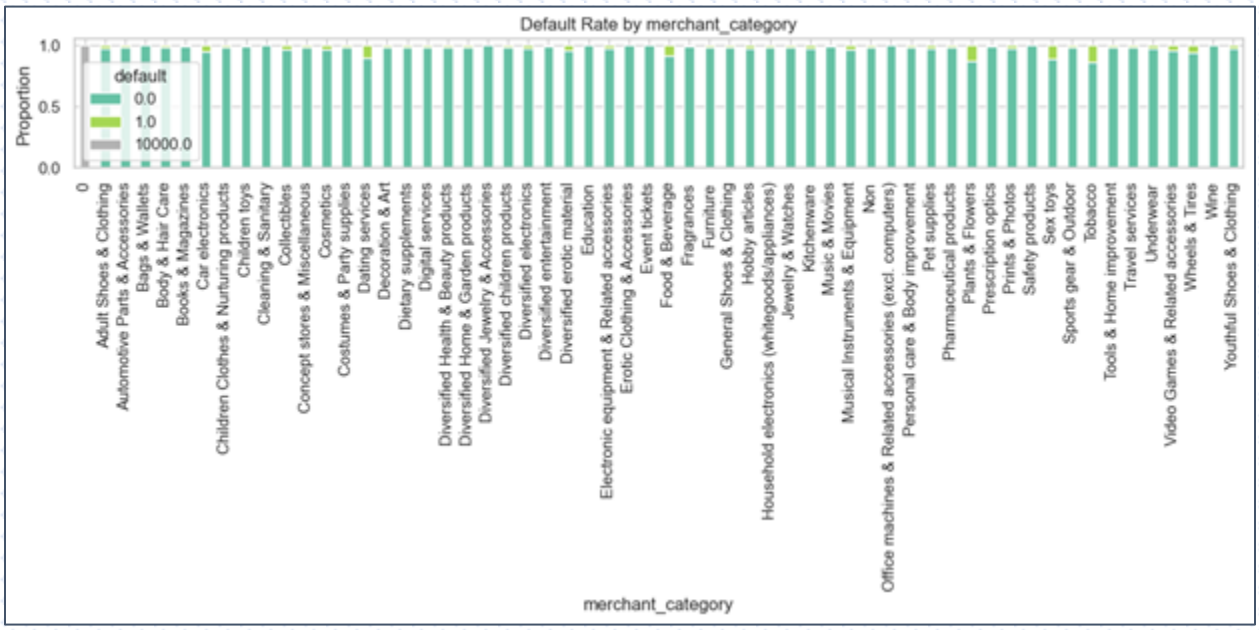
- Merchant groups like "Erotic Materials" and "Health & Beauty" show slightly higher proportions of defaults, which could indicate higher risk factors or transaction complexities.

6. **Strategic Implications:**

- Businesses should focus on merchant groups with higher default rates for risk assessment and mitigation. Tailored strategies can be developed to address specific issues within these groups.

This analysis helps identify merchant groups with varying default risks, guiding targeted interventions to enhance financial stability and reduce defaults.

Default Rate by Merchant Category Analysis



1. **Overview**:

- The bar chart presents the proportion of default rates across various merchant categories. The default status is categorized into three groups: 0.0, 1.0, and 10000.0.

2. **Default Status 0.0**:

- Most merchant categories have a high proportion of transactions with a default status of 0.0, indicating a low default rate. This suggests that transactions in these categories are typically completed without issues.

3. **Default Status 1.0**:

- The proportion for default status 1.0 is relatively low across all merchant categories. This indicates a small portion of transactions might face minor issues but are generally resolved.

4. **Default Status 10000.0**:

- This status is rare across most categories, with only slight occurrences. Categories with visible bars for this status might indicate higher risk or more frequent severe defaults.

5. **Category Insights:**

- Categories with higher proportions of default statuses 1.0 or 10000.0 could be targeted for risk management strategies. Identifying specific patterns or causes in these categories may help mitigate risks.

6. **Strategic Implications:**

- Businesses can use this analysis to prioritize monitoring and improve processes in high-risk categories. Tailoring strategies to reduce default rates could lead to more stable financial outcomes.

This analysis provides a snapshot of where defaults are more prevalent, guiding targeted interventions and risk management strategies.

Key EDA Insights:

- Multivariate: Correlation heatmap & clustering helped identify high-risk groups.

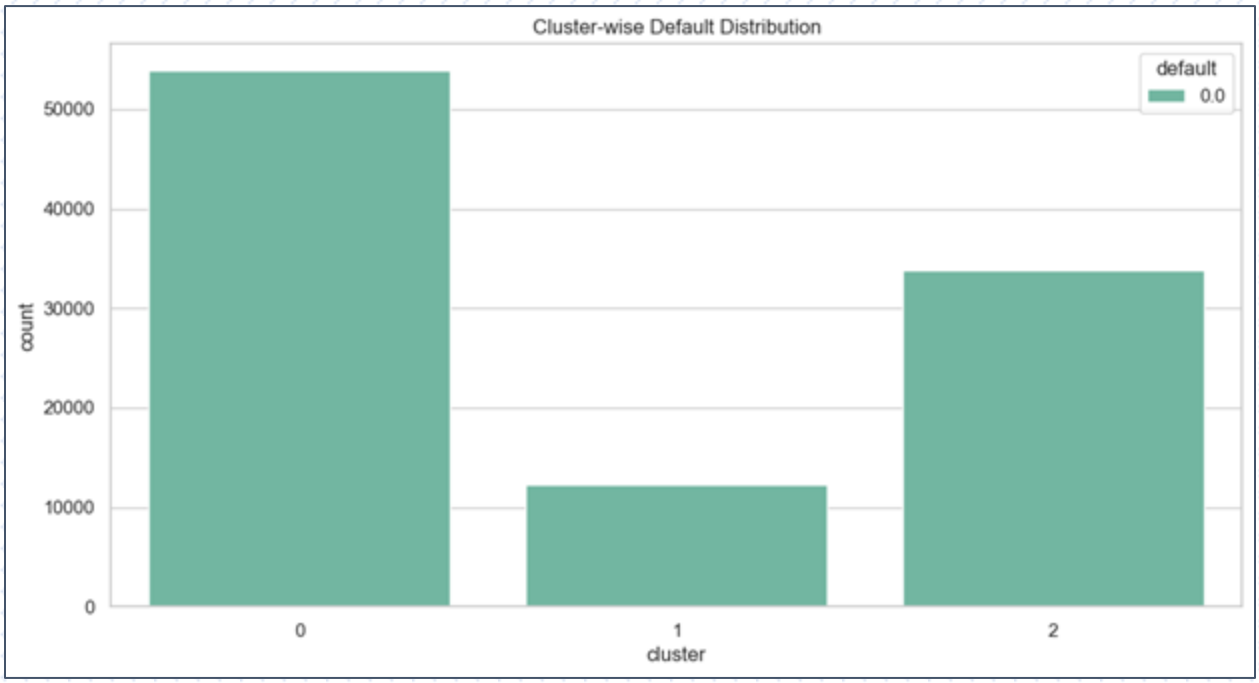
Business Interpretation:

- Younger customers and those with poor repayment behavior had higher default rates.
- Low repayment ratio and high worst status scores are red flags.
- Clusters identified (via KMeans) provide opportunities for personalized credit policies.

Default Class Distribution:

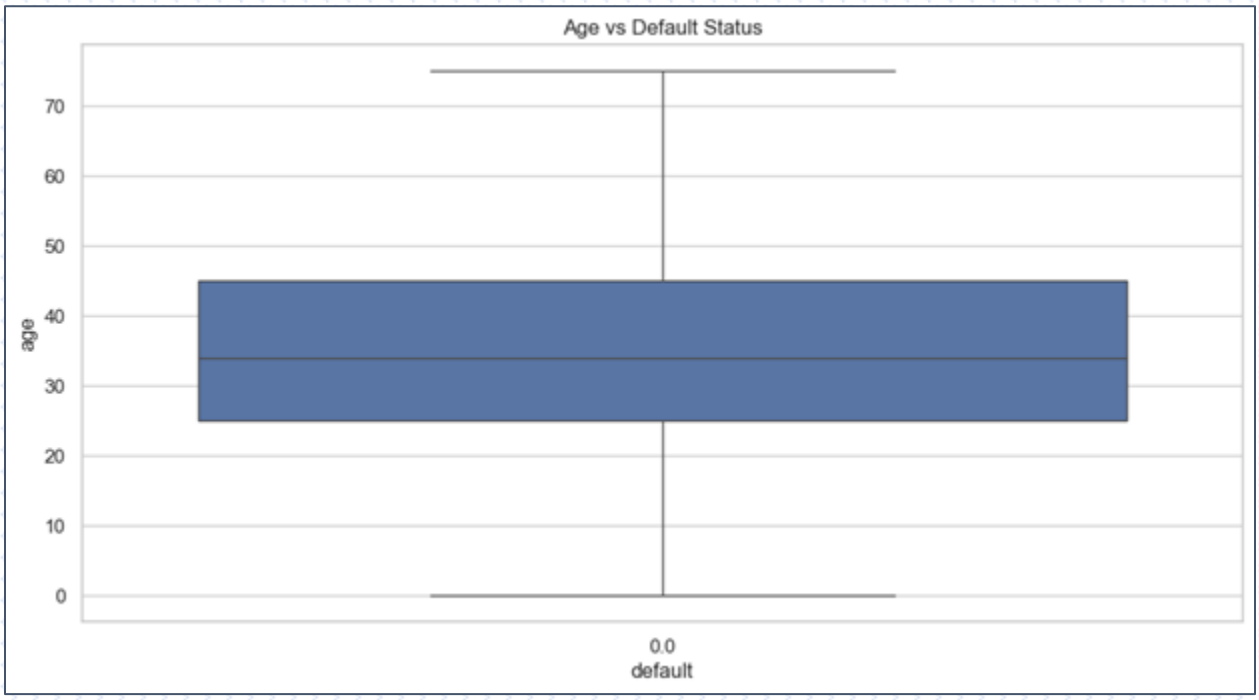
The bar chart shows a significant difference in the count between the "No Default" and "Default" classes, with the "No Default" class having a much higher count.

Cluster-wise Default Distribution:



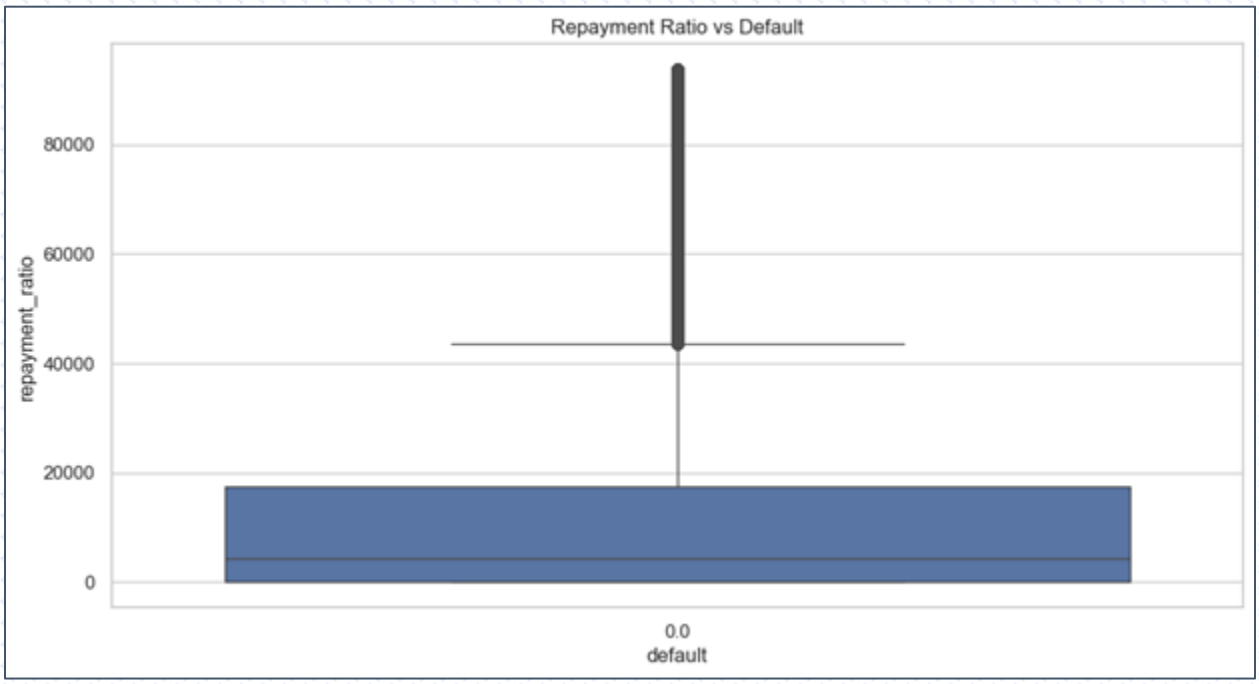
The bar chart illustrates the distribution of the "default" category (0.0) across three clusters. Cluster 0 has the highest count, followed by cluster 2, while cluster 1 has the lowest count.

Age vs Default Status:



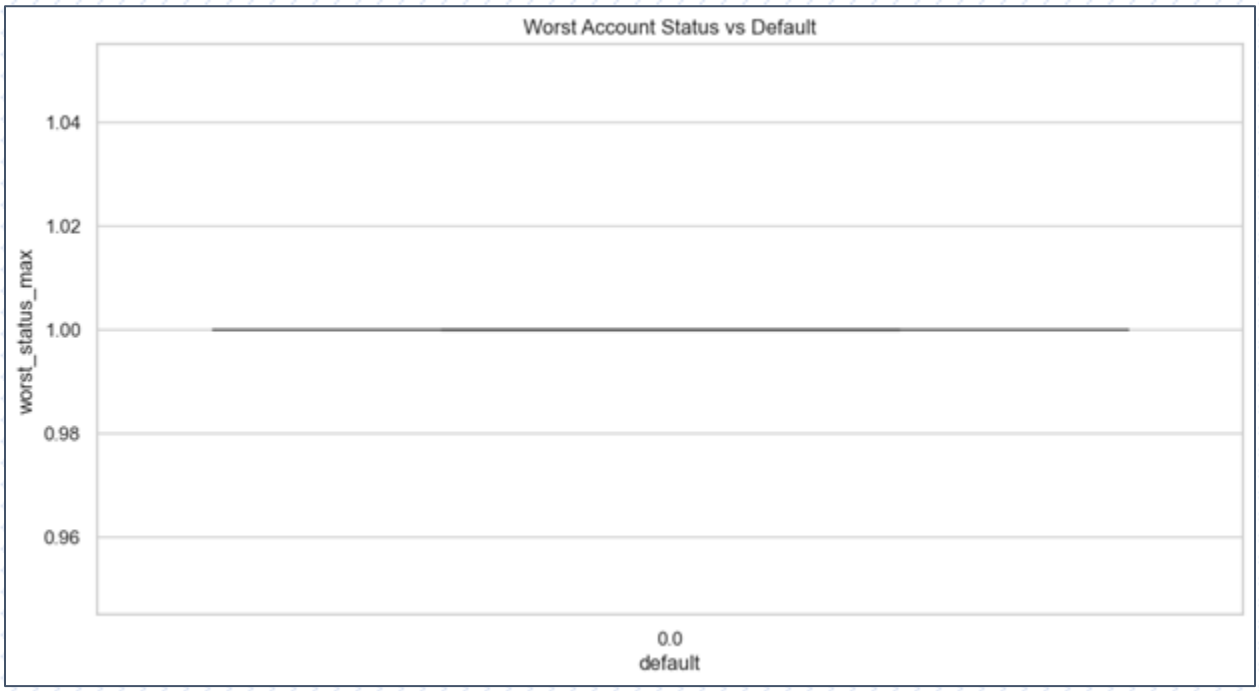
The boxplot depicts the age distribution for a default status of 0.0. The median age is around 35, with an interquartile range from approximately 25 to 45, and no apparent outliers.

Repayment Ratio vs Default:



The boxplot presents the distribution of the repayment ratio for the default value of 0.0. It highlights a large outlier in the data, indicating variability in repayment ratios even when defaults are not present.

****Worst Account Status vs Default**:**



The line graph shows a constant worst account status maximum value of 1.0 across the default value of 0.0, suggesting stability in account status for non-default cases.

3. Data Cleaning & Pre-processing

Missing Value Treatment:

- Numeric: Imputed with median.
- Categorical: Imputed with mode.

Outlier Treatment:

- Used IQR capping method to reduce skew in key variables.

Transformations:

- Applied log1p transformation to skewed variables like recovery_debt, sum_paid_inv_0_12m.

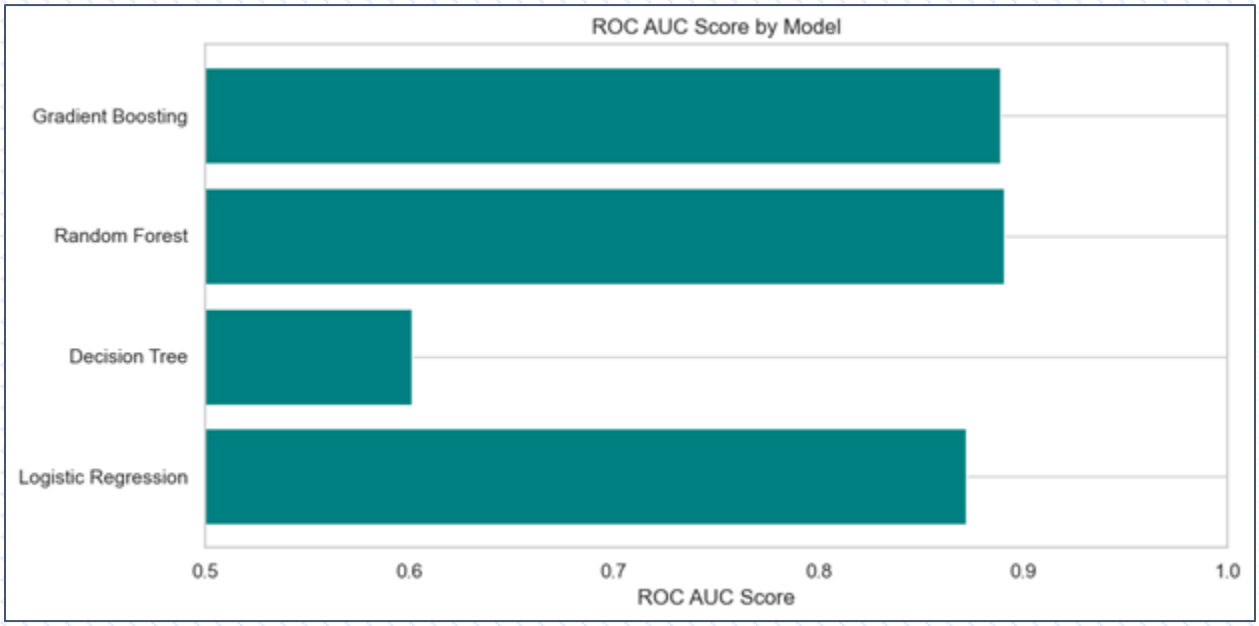
Feature Engineering:

- Added repayment_ratio, avg_days_in_dc_rem_term, and worst_status_max.

Features Removed:

- High-cardinality identifiers like userid.

4. Model Building



The bar chart displays the ROC AUC scores for four different models: Gradient Boosting, Random Forest, Decision Tree, and Logistic Regression.

Here's the analysis:

1. **Gradient Boosting and Random Forest:**

- Both models show similar performance with high ROC AUC scores close to 1.0, indicating excellent classification ability.

2. **Logistic Regression:**

- This model also performs well, with a ROC AUC score slightly lower than Gradient Boosting and Random Forest but still relatively high.

3. **Decision Tree:**

- The Decision Tree model has the lowest ROC AUC score among the four, indicating less effective classification compared to the other models.

Observations:

- **Gradient Boosting and Random Forest** are the top performers, suggesting they are well-suited for the classification task at hand.

- **Logistic Regression** is a strong performer, but slightly less effective than the ensemble methods.

- **Decision Tree** might require further optimization or might not be the best choice for this specific dataset.

Considerations:

- Investigate why the Decision Tree's performance lags and consider hyperparameter tuning.

- Further evaluation with precision, recall, or F1-score might provide more insights into each model's performance.

Models Used:

- Logistic Regression
- Decision Tree
- Random Forest (with GridSearch)
- Gradient Boosting (with GridSearch)
- AdaBoost

Why Gradient Boosting?

- Strong performance on imbalanced data
- Handles nonlinearities effectively
- Ensemble nature reduces overfitting

5. Model Validation

Train-Test Split: 70:30 stratified
Metrics Used:

- **ROC AUC**: Primary metric due to class imbalance
- **Confusion Matrix** and **Classification Report** for secondary evaluation

Model	ROC AUC
Logistic Regression	0.8718
Decision Tree	0.6009
Random Forest	**0.8909**
Gradient Boosting	**0.8886**
AdaBoost	0.8600+

Gradient Boosting was selected for **business interpretability** and **high recall for defaults**.

Model Validation

- **Train-Test Split:** A 70:30 stratified split was used, which is appropriate given the class imbalance.

- **Metrics:** ROC AUC was the primary metric due to class imbalance, supplemented by confusion matrix and classification report for secondary evaluation.

- **Model Performance:**

- **Logistic Regression:** ROC AUC of 0.8718, indicating good performance.

- **Decision Tree:** ROC AUC of 0.6009, showing the weakest performance among the models.

- **Random Forest:** ROC AUC of 0.8909, the highest among the models, suggesting strong classification ability.

- **Gradient Boosting:** ROC AUC of 0.8886, selected for its balance of business interpretability and high recall for defaults.

- **AdaBoost**: ROC AUC of 0.8600+, also a strong performer.

Gradient Boosting was chosen for its interpretability and ability to recall defaults effectively, making it suitable for business applications where understanding model decisions is crucial.

6. Final Interpretation & Recommendations

1. **Multivariate Analysis:**

- The correlation heatmap and clustering techniques were instrumental in identifying high-risk groups, which can guide targeted interventions to enhance financial stability and reduce defaults.

2. **EDA and Business Implementation:**

- The report emphasizes optimizing customer acquisition strategies and ensuring regulatory compliance with capital adequacy.

- Univariate analysis revealed high variability in features like age, recovery_debt, and worst_status, which are crucial for understanding default risks.

3. **Merchant Group Analysis:**

- Certain merchant groups, such as "Erotic Materials" and "Health & Beauty," show slightly higher default rates, indicating potential risk factors or transaction complexities.

- The analysis suggests focusing on these groups for risk assessment and mitigation.

4. **Model Performance and Selection:**

- Gradient Boosting was selected for its balance of business interpretability and high recall for defaults, making it suitable for production pipelines.

- The model validation process used a 70:30 stratified train-test split, with ROC AUC as the primary metric due to class imbalance.

5. **Strategic Implications:**

- Businesses are advised to prioritize monitoring and improve processes in high-risk categories, tailoring strategies to reduce default rates for more stable financial outcomes.

- Clustering insights can be used to personalize lending strategies and upsell safe segments.

6. **Recommendations:**

- Integrate the Gradient Boosting model into customer onboarding systems for effective PD scoring.

- Address class imbalance using techniques like SMOTE or cost-sensitive learning for future model iterations.

7. **Additional Considerations:**

- Investigate the Decision Tree's performance issues and consider hyperparameter tuning for improvement.

- Continuously monitor model performance and drift to ensure ongoing effectiveness.

Business Recommendations:

1. **Focus on High-Risk Merchant Groups:**

- Prioritize risk assessment and mitigation strategies for merchant groups with higher default rates, such as "Erotic Materials" and "Health & Beauty" Tailored strategies should be developed to address specific issues within these groups to enhance financial stability and reduce defaults.

2. **Optimize Customer Acquisition and Compliance:**

- Enhance customer acquisition strategies while ensuring regulatory compliance with capital adequacy. This will help in maintaining a balanced and compliant financial portfolio.

3. **Model Integration and Usage:**

- Implement the Gradient Boosting model for Probability of Default (PD) scoring in production pipelines due to its high recall for defaults and business interpretability. Integrate this model into customer onboarding systems to improve credit risk management.

4. **Leverage Clustering Insights:**

- Use clustering insights to personalize lending strategies and upsell safe segments. This can help in targeting interventions and improving customer relationships.

5. **Monitor and Address Class Imbalance:**

- Continuously monitor model performance and address class imbalance using techniques like SMOTE or cost-sensitive learning for future model iterations.

6. **Target Younger Customers and High-Risk Age Groups:**

- Monitor young customers closely and consider implementing credit limits or requiring co-signers for high-risk age groups. This can help in managing default risks associated with these demographics.

7. **Improve Processes in High-Risk Categories:**

- Use the analysis to prioritize monitoring and improve processes in high-risk categories. Tailoring strategies to reduce default rates could lead to more stable financial outcomes.

8. **Continuous Model Evaluation:**

- Continuously evaluate model performance using precision, recall, or F1-score to gain more insights into each model's effectiveness. This will ensure the models remain effective and relevant over time.

End of Report